INCH-POUND

MIL-D-45099E(AL)
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SUPERSEDING
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MILITARY SPECIFICATION

DEGREASERS, STATIONARY, HIGH VELOCITY, CENTRIFUGAL IMPACT TYPE, OIL FIRED

This specification is approved for use by the Armament Research and Development Engineering Center (Tool and Equipment Engineering Division, Standardization and Specification Engineering Branch), Department of the Army and is available for usage by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers centrifugal, impact-type degreasers using hot alkali or cold petroleum solvents as the cleaning solution.

1.2 Classification. The degreasers shall be of the following sizes, as specified (see 6.2.1):

Size 1	-	100 gallon minimum capacity
Size 2	-	300 gallon minimum capacity

2. APPLICABLE DOCUMENTS

*2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation. (see 6.2.1)

SPECIFICATIONS

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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O-A-548 - Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited, Concentrated.
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular, Polypropylene Foam.
PPP-T-60 - Tape, Packaging, Waterproof.
PPP-T-97 - Tape, Pressure Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-C-104 - Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-C-14460 - Corrosion Removing Compound, Sodium Hydroxide Base; For Electrolytic or Immersion Application.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

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(Copies of specifications, standards, and other Government drawings and publications required by contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

*2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of the documents cited in the solicitation. (see 6.2.1)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.
AGMA 390.03 - Gear Handbook, Volume 1, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Control and Systems.

NEMA MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

*2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The degreasers shall be new and of the manufacturer's current design meeting the requirements of this specification. Degreasers shall be provided with dual impeller agitation, shall be oil fired, and be capable of cleaning engine blocks and other metal parts. The degreasers shall be capable of removing heavy deposits of sludge, oil, metal chips, compounds, varnish, and paint from metal objects using sodium hydroxide base compounds (see 3.9.2). The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the degreasers or any component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the degreasers. In this specification, all measurements, dimensions, sizes and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380.

3.2.2 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

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3.2.3 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.4 Gears. All gears and pinion designs selected for use in the degreaser or its components shall be manufactured to either English (USA) standard inch diametral pitch in accordance with AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio in accordance with International Organization for Standardization (ISO) ISO 54 and ISO 1122/1 and ASTM E380.

3.2.5 Safety and health requirements. Covers, guards, and other safety devices shall be provided. The devices shall not interfere with operation of the degreaser. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of parts. Exceptions and addional requirements for safety and health shall be as specified in the contract (see 6.2.1).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The degreasers shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding, without distortion, all forces encountered during operation of the degreasers at maximum rating and capacity.

*3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in

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accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.4 Lubrication. All bearings (except sealed, permanently lubricated type), mating gears, and other moving parts shall be provided means to assure adequate lubrication. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. Oil holes, grease fittings, and filler caps shall be located to be accessible. The manufacturer shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practice of the producing industry (see 6.4). The lubrication systems shall be temporarily capped or stopped to prevent loss of lubricants during shipment. The manufacturer shall include a warning notice in the instruction manual and on the degreasers to remove the temporary caps or stoppers before start-up or use of the degreasers.

3.4.4.1 Lubrication chart or plate. A corrosion-resistant metallic lubrication plate, with permanently legible characters in the English language, shall be attached to each degreaser in a location visible to the operator. If a lubrication chart is furnished, it shall be placed in a transparent plastic folder or laminated between clear plastic permanently sealed sheets with suitable means for mounting. The following information (see 6.4) shall be furnished on the chart and plate:

Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal specification number of lubricant.

3.4.5 Surfaces. Surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, flash, and other harmful or extraneous materials.

3.4.6 Protective finishes. Exposed metal (see 6.5.1) and working surfaces, except when the parts involved are inherently corrosion resistant, shall be protected with an electrochemical or chemical conversion finish.

3.4.7 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in soldering operations.

3.5 Performance and product characteristics.

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3.5.1 Capacity. The Size 1 degreaser shall have a liquid capacity of no less than 100 gallons and a work basket holding capacity of no less than 500 pounds. The Size 2 degreaser shall have a liquid capacity of no less than 300 gallons and a work basket holding capacity of no less than 2000 pounds.

3.5.2 Burner and combustion chamber performance. The degreasers shall be capable of raising the temperature of the cleaning solution from +70 DEG. Fahrenheit (F) +10 DEG. F to a minimum of +200 DEG. F within two hours for the Size 1, and three hours for the Size 2, from the time the burner is ignited. The degreasers shall be capable of maintaining a temperature between +190 DEG. F and +200 DEG. F on a maximum consumption of 1 1/2 gallons for the Size 1 and three gallons for the Size 2 of No. 2 grade diesel fuel oil per hour. The burners shall be regulated to provide continuous, complete combustion of the fuel oil during the initial heating phase and also during maintenance of the normal operating temperature range. The degreasers shall be capable of regaining the solution operating temperature after being unloaded and reloaded with parts that are at room temperature. The maximum time required for the solution to regain the drop in temperature, occurring after each reloading of metal parts or engine blocks, shall not exceed ten minutes.

3.5.3 Continuous operation. After being operated continuously for a period of 40 hours, while loaded to capacity with cleaning solution and a maximum capacity load of metal parts or engine blocks, the degreasers shall show no evidence of leakage, failure, or damage to any component. The temperature of the cleaning solution shall be maintained between +190 DEG F and +200 DEG F during this operation.

3.5.4 Overflow, drain, and sediment facilities. Each degreaser shall have provision for the overflow and drainage of excess solution. The overflow shall be capable of handling the solution during the most violent agitation at full capacity. That is, the solution shall not spill or splash over the sides of the tank. The drain and overflow shall be interconnected and shall have control valves accessible to the operator. The tank shall be constructed to provide a settling basin for the sediment. The settling basin shall prevent agitation of the sediment and the remixing of it with the solution while the impellers are operating. Provision shall be made for periodic emptying, flushing, or draining of the settling basin and sediment tank.

3.5.5 Controls. Controls shall be provided and mounted on the degreasers for selecting and automatically maintaining the solution temperature in addition to necessary motor and burner controls. All controls shall be accessible to the operator.

3.6 Components.

3.6.1 Tank and tank cover. The tank shall be fabricated from steel not less than Manufacturers' Standard Gage No. 10 (0.135 inch) and reinforced with structural steel throughout. The cover shall be fabricated from steel not less

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than Manufacturers' Standard Gage No. 16 (0.060 inch). The cover shall open fully to allow for the vertical insertion and removal of the work basket or any parts that are being cleaned. The tank rim shall be constructed so that solution dripping from parts being removed shall drain back into the tank. The tank shall also be so designed and constructed so there is no surface or area, except for the sediment basin, where effective agitation does not reach. Safety step(s) with a nonskid surface shall be provided at the front of the degreasers when the height of the tank exceeds four feet.

3.6.2 Settling basin (tank). The settling basin (tank) shall be provided with means such as a pan, trap, or drain for the collection and removal of sediment. Sediment removal shall be capable of being performed, when the tank is filled to normal capacity, with loss of not more than five percent of the cleaning solution.

3.6.3 Work basket support. A steel grating or support(s) shall be provided in the tank to support the work basket and shall be located to provide for free flow of the solution to, and around, the parts being cleaned. The location shall prevent contact with any of the moving parts of the degreasers. The grating or support(s) shall be capable of supporting the work load specified in 3.5.1 without evidence of damage or deformation.

3.6.4 Work basket. A steel work basket shall be supplied with each degreaser. The basket shall be capable of holding a uniformly distributed load, as specified in 3.5.1, without evidence of damage or deformation. The basket shall be constructed to permit the flow of solution through all areas. The basket shall have a lifting handle(s).

3.6.5 Overflow and drain. The overflow facility shall be constructed to carry off excess solution before it can overflow the tank.

3.6.6 Water intake line and control. The water intake line shall be supplied with a valve to control the tank's water supply. This valve shall be located above the overflow facility to prevent possible contamination of the water intake line by the cleaning solution. The valve shall be accessible to the operator.

3.6.7 Impellers. The Size 1 degreasers shall have a minimum of one impeller, driven by a motor of two horsepower (hp) minimum. The Size 2 degreasers shall have a minimum of 2 impellers, each driven by a motor of five hp minimum. The impellers shall be fabricated from stainless steel or shall be stainless steel castings. The impellers shall be a minimum of ten inches in diameter and shall have a minimum of three vanes each. The impeller shaft shall be mounted in antifriction-type bearings. The bearings and shaft shall be provided with seals or fluid impervious packing to insure a positive seal against the entry of water or solution. The impellers shall be mounted to effectively agitate the solution, but not interfere with the operation or loading and unloading of the tank.

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3.6.8 Oil burner. The oil burner shall be a type approved by the Underwriters' Laboratories and be of the horizontal rotary cup atomizing type, or gun type, with blower, ignition system, and fuel pump integrally mounted. The burner shall be capable of performance in accordance with 3.5.2. The oil burner shall be capable of operating on No. 2 grade diesel fuel oil. The oil burner motor shall be rated at not less than 1/8 hp, 115 volt, 60 hertz (Hz), single phase. The oil burner shall be equipped with automatic ignition and automatic controls.

3.6.9 Combustion chamber and exhaust stack. The combustion chamber or fire box shall be fabricated of stainless steel, low carbon steel with stainless steel lining, or low carbon steel with fire brick lining. Means shall be provided for inspection and cleanout. The combustion chamber shall be located to provide maximum heat transfer to the solution in the tank. The exhaust stack shall be a minimum of six inches in diameter for Size 1 degreasers and a minimum of eight inches in diameter for Size 2 degreasers.

3.6.10 Temperature and safety controls. The temperature of the cleaning solution shall be maintained automatically and by preselection, with provision to automatically shut down the burner in the event of overheating. Overheating shall not exceed +255 DEG. F. A thermometer or temperature gage shall be provided with a range from not more than +50 DEG. F to not less than +300 DEG. F. Each degreaser shall be provided with Underwriters Laboratories approved safety controls that shall automatically shut off the fuel supply in the event of flame or ignition failure, or the simultaneous failure of both.

3.7 Electrical equipment.

3.7.1 Motors. Motors shall be induction squirrel cage type conforming to NEMA MG1, with class B insulation and continuous duty rating. Unless otherwise specified (see 6.2.1), the impeller motors shall be dual voltage, 230/460 volts, initially wired for operation at 230 volts, 60 Hz, three phase.

3.7.2 Electrical components. Each degreaser shall be completely wired for the operations specified herein. All electrical components of the degreasers including starters, relays, switches, wiring, and their inclosures shall conform to, and be located in accordance with, NEMA ICS 1 and NFPA 70, as applicable. Components shall not be used in any application where the manufacturer's rating for the part does not meet or exceed the requirements.

3.7.3 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

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3.8 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.9 Product identification.

3.9.1 Nameplate. The nameplate and information plate(s) may be incorporated into one or more corrosion-resistant metallic plate(s) at the manufacturer's option. The plate(s) shall conform to MIL-STD-130. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The plate(s) shall be secured to the degreasers with permanent fasteners at each of the four corners. The plate(s) shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number

3.9.1.1 Information plate. When applicable (see 3.9.1), a corrosion resistant metallic information plate, with permanently legible characters in the English language, shall be permanently fastened to each degreaser in a location visible to the operator. Characters and numerals shall be engraved, etched, embossed, or stamped in bold face on a contrasting background. The information plate shall be secured to the degreasers with permanent fasteners at each of the four corners. The information plate shall include warnings, cautions, and operator instructions to be followed when operating or maintaining the degreaser.

3.9.2 Warning plate. A corrosion resistant metallic warning plate shall be permanently attached to the degreaser in a location where it shall be in full view of the operator. The printing shall be in permanent, bold letters not less than 3/4 inch high for the word "WARNING" and 1/4 inch high for all other words. The warning shall be as follows:

WARNING

USE ONLY SODIUM HYDROXIDE BASE CORROSION REMOVING COMPOUND CONFORMING TO MIL-C-14460, TYPE I, NSN 6850-00-550-5565 (400 POUND DRUM). THIS DEGREASER, IF USED IN A CONFINED AREA, SHALL BE PROVIDED WITH A FORCED VENTILATION SYSTEM.

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3.10 Workmanship. Standards of workmanship shall assure that the degreasers shall have the stability, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

*4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (d) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Performance test (see 4.6.1).
(c) Flame failure test (see 4.6.3).
(d) Packaging inspection (see 4.7).

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*4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or test	Lot or batch size	Sample size
(a) Product examination (4.5)	1-150	13
(b) Performance test (4.6.1)	1-150	13
(c) Flame fialure test (4.6.3)	1-150	13
(d) Packaging inspection (4.7)	1-150	5

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100% inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

*4.5 Product Examination. Visually, dimensionally, and manually examine each degreaser to determine conformance with the requirements of 3.2 through 3.4.7, and 3.6 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards and adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacturing at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operation. Failure of the contractor to provide objective evidence that the degreaser and it's components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.2.2), shall be cause for rejection.

4.6 Tests.

4.6.1 Performance test. Each degreaser shall be loaded to rated capacity with a cleaning solution and the burner (see 3.6.8) shall be ignited to determine compliance with the heat-up time specified in 3.5.2. When the temperature of the solution has reached a minimum of +200 DEG. F, the degreaser shall be loaded to capacity, as specified in 3.5.1, with metal parts or engine blocks that have been stabilized at room temperature. The time required to regain the temperature drop shall not exceed ten minutes. After the degreaser has reached the required temperature, the impeller motor(s) shall be turned on to assure proper operation and presence of sufficient agitation to effect cleaning action. The grating support shall not show any evidence of damage or deformation upon completion of the test (see 3.6.3).

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*4.6.2 Continuous operation. Each degreaser shall be loaded to the rated capacity with the manufacturer's recommended cleaning solution and the weight of metal parts or engine blocks specified in 3.5.1. The degreaser shall be operated continuously for a minimum of 40 hours to determine compliance with 3.5.3. The cleaning solution shall be maintained at a temperature between +190 DEG F and +200 DEG F during this operation. The fuel consumption shall be checked to determine conformance with 3.5.2. During and following this test, the degreaser shall show no evidence of leakage or failure or damage to any component. Upon completion of this test, the degreaser shall be subjected to the tests specified in 4.6.3 and 4.6.4.

*4.6.3 Flame failure test. While each degreaser is in full operation, a flame failure shall be simulated to determine if the safety controls have automatically stopped fuel flow as specified in 3.6.10. The burner shall be restarted and any evidence of excessive fire or explosion which causes damage to the degreaser shall be cause for rejection.

*4.6.4 Ignition failure test. Each degreaser shall be started with the ignition disconnected. The fuel shall automatically shut off. The ignition shall be reconnected and, with the degreaser in full operation, an ignition failure shall be simulated to determine if the safety controls have automatically stopped fuel flow as specified in 3.6.10. The burner shall be restarted and any evidence of explosion which causes damage to the degreaser shall be cause for rejection.

4.6.5 Cleaning solution overheating test. The preselective heat range operating controls shall be disconnected to allow overheating of the cleaning solution. When the solution has been heated to exceed the temperature of +255 DEG F, operation of the safety controls shall shut down the burner (see 3.6.10).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Disassembly. The degreasers may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

5.2 Preservation. Preservation shall be Level A or Level C as specified (see 6.2.1).

5.2.1 Level A.

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5.2.1.1 Cleaning and drying. The degreaser shall be cleaned in accordance with process C-1 and dried in accordance with any applicable procedures of MIL-P-116.

5.2.1.2 Preservative application. Immediately after cleaning and drying, apply a coat of type P-1 or type P-19 preservative to all bare, exterior, ferrous metal surfaces of the degreaser and accessories. Interior surfaces of the fuel system shall be preserved with type P-10, grade 10 preservative oil. Interior surfaces of the water system shall be flushed with a solution consisting of 50 percent ethylene glycol of O-A-548 and 50 percent clean fresh water. After flushing, the system shall be thoroughly drained. Preservatives specified and the procedures for application shall conform to MIL-P-116.

5.2.1.3 Unit protection. The degreaser shall be preserved in accordance with method I of MIL-P-116.

5.2.1.3.1 Electrical components. The electric motor(s) shall be processed in accordance with Level A requirements of MIL-E-16298. Exposed ends of electric wires, plug openings, sockets, terminals, and openings into switch boxes shall be sealed with tape conforming to PPP-T-60, type IV, or caps or plugs of MIL-C-5501. Electric cables shall be coiled to a minimum safe diameter and secured with cotton tape or twine to prevent uncoiling.

5.2.1.3.2 Controls and gages. Temperature and safety controls, gages, and electric switches shall be covered with 1/2 inch thick cushioning material conforming to PPP-C-1797. The material shall be secured in place with tape conforming to PPP-T-60, type optional.

5.2.1.3.3 Work basket. When feasible, the work basket shall be secured to the work basket supports with common annealed wire or steel strapping.

5.2.1.3.4 Water and oil intake lines. Water and oil intake lines shall be sealed with the tape or cap or plugs specified in 5.2.1.3.1.

5.2.1.3.5 Exhaust stack. The exhaust stack shall be packaged in a closefitting box conforming to PPP-B-636, class domestic, variety and grade optional. Secure the container to the work basket with flat steel strapping.

5.2.1.3.6 Technical data. When applicable, publications issued with the degreaser shall be packaged in accordance with method IC-1 of MIL-P-116.

5.2.1.3.7 Bearings. Bearings, except sealed-for-life type, shall be charged with type P-11 preservative of MIL-P-116.

5.2.1.3.8 Drive belts. Drive belts shall be removed or released from tension. The faces and grooves of ferrous metal pulleys shall be coated with primer conforming to TT-P-664.

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5.2.1.3.9 Fire brick. Refactory linings, when applicable, shall be protected against damage.

5.2.1.3.10 Consolidated pack. Technical data, parts removed thru disassembly, and accessories, except for the safety steps and fire brick, shall be packaged in a box(s) conforming to PPP-B-636, class domestic, variety and grade optional, or PPP-B-621, class 1. The boxes shall be secured to the work basket with flat steel strapping as referenced in the box specifications.

5.2.1.3.11 Safety steps. Space permitting, the safety steps shall be placed inside the degreaser and secured to the work basket with flat steel strapping or common annealed wire. Otherwise, the steps shall be cushioned and secured on top or along the side of the degreaser, inside the shipping container.

5.2.1.3.12 Tank cover. The tank cover shall be secured in the closed position using filament tape conforming to type III of PPP-T-97 or flat steel strapping (see 5.2.1.3.10). When filament tape is used, it shall be wrapped around the entire tank, in two places, with a minimum six inch overlap. When steel strapping is used, edge protectors shall be used to protect the degreaser from damage. The tape or strapping shall be located approximately 1/3 of the tank length in from each end of the tank cover.

5.2.2 Level C. Level C preservation shall be as specified in MIL-STD-1190.

5.3 Packing. Packing shall be Level A or Level C as specified (see 6.2.1).

5.3.1 Level A. Each degreaser shall be packed in a crate conforming to MIL-C-104, type I, class 2, style a. Arrangement of contents, anchoring, blocking, bracing, and cushioning shall conform to MIL-STD-1186.

5.3.2 Level C. Level C packing shall be as specified in MIL-STD-1190.

5.4 Marking. All marking requirements shall be in accordance with MIL-STD-129.

*5.5 Rough handling. The completed unit pack, shipping container(s) and all contents shall capable of withstanding the shipping and handling conditions expected from fork lift handling and truck transport. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test(s) specified elsewhere in the contract.

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6. NOTES

6.1 Intended use. The degreasers covered by this specification are intended for use in engine rebuild and repair shops for the removal of grease, sludge, and other foreign materials from vehicle engine blocks and parts. The cleaning action is effected by agitating a heated solution or cold petroleum base cleaning solution over the objects being cleaned.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
(c) Size required (see 1.2).
(d) First article when required (see 3.1).
(e) Exceptions and additional safety/health requirements when required (see 3.2.5).
(f) Impeller motors if different (see 3.7.1).
(g) Painting and finishing if different (see 3.8).
(h) First article inspection if different (see 4.3).
(i) Quality conformance inspection if different (see 4.4).
(j) Level of preservation, packing, and marking required (see 5.2, 5.3, and 5.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.4 Standard lubricants. The degreaser manufacturer should specify and use to the maximum extent possible those lubricants (see 3.4.4) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

6.5 Definitions.

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6.5.1 Exposed metal. Exposed metal is defined as metal exposed to atmospheric conditions, cleaning solvents, and oils and greases (see 3.4.6).

*6.6 Subject term (key word listing.

Centrifugal
Oil fired
Impeller
Work basket

6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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